Hypothesis Testing (continued) Exercises Solutions

Exercise 7.1. *A production line is designed to fill bottles with oil. The amount of oil placed in a bottle is normally distributed and the mean is set to 100 ml. The amount of oil, $x$ml, in each of 8 randomly selected bottles is recorded, and the following statistics are obtained.*

$$\bar{x} = 92.875, \sigma_x = 8.3055$$

The supervisor believes that the mean amount of oil placed in a bottle is less than 100 ml. Stating your hypotheses clearly, test, at the 5% significance level, whether or not their belief is supported.

$H_0 : \mu = 100$
$H_1 : \mu < 100$
Perform a one-sample t-test with $\nu = 8 - 1 = 7$. The test statistic is $t = \frac{\bar{x}-\mu}{\frac{\sigma_x}{\sqrt{n}}} = \frac{92.875-100}{\frac{8.3055}{\sqrt{8}}} = -2.426$
The critical region for $\nu = 7$ is -1.895 or lower. Since t is in the critical region, the result is significant, and we can reject the null hypothesis that the mean amount of oil in the bottles is still 100ml.

Exercise 7.2. *(a) Two methods of extracting juice from an orange are to be compared. Eight oranges are halved. One half of each orange is chosen at random and allocated to Method A and the other half is allocated to Method B. The amounts of juice extracted, in ml, are given in the table.*

Orange	*1*	*2*	*3*	*4*	*5*	*6*	*7*	*8*
Method A	*29*	*30*	*26*	*25*	*26*	*22*	*23*	*28*
Method B	*27*	*25*	*28*	*24*	*23*	*26*	*22*	*25*

One statistician suggests performing a two-sample t-test to investigate whether or not there is a difference between the mean amounts of juice extracted by the two methods. Another statistician suggests analysing these data using a paired t-test.

(i) Explain why the paired t-test is a more appropriate test to use.
(ii) Carry out this test.

(i) The data are paired: each orange has a pair of values assigned to it.
(ii) $H_0 : \mu_D = 0$
$H_1 : \mu_D \neq 0$, *where* $D = X_A - X_B$
Perform a two-tailed paired t-test with $\nu = 8 - 1 - 7$.
The mean of the differences is

$$\frac{(29-27)+(30-25)+(26-28)+(25-24)+(26-23)+(22-26)+(23-22)+(28-25)}{8} = \frac{9}{8} = 1.125$$

The standard deviation is 2.90. *Then the test statistic is* $t = \frac{\bar{d}-\mu}{\frac{\sigma}{\sqrt{8}}} = \frac{1.125-0}{1.025} = 1.097$
The critical region for $\nu = 7$ is -2.365 and below OR 2.365 and above. Since 1.097 is not in the critical region, this result is not significant, and there is insufficient evidence to reject the null hypothesis that there is a difference in the amount of juice.

(b) As part of an investigation into the effectiveness of solar heating, a pair of houses was identified which had the same mean weekly fuel consumption. One of the houses was then fitted with solar heating and the other

was not. Following the fitting of the solar heating, a random sample of 9 weeks was taken and the table below shows the weekly fuel consumption for each house.

Week	1	2	3	4	5	6	7	8	9
Without solar heating	19	19	18	14	6	7	5	31	43
With solar heating	13	22	11	16	14	1	0	20	38

(i) Test whether or not there is evidence that the solar heating reduces the mean weekly fuel consumption.

(ii) State an assumption about weekly fuel consumption that is required to carry out this test.

(i) $H_0 : \mu_D = 0$
$H_1 : \mu_D > 0$, where $D = X_{without} - X_{with}$.
Perform a one-tail paired t-test with $\nu = 9 - 1 - 8$. The mean of the differences is

$$\frac{(19-13)+(19-22)+(18-11)+(14-16)+(6-14)+(7-1)+(5-0)+(31-20)+(43-38)}{9} = \frac{27}{9} = 3$$

The standard deviation is 6. Then the test statistic is $t = \frac{\bar{d}-\mu}{\frac{\sigma}{\sqrt{9}}} = \frac{3-0}{2} = 1.5$
The critical region for for $\nu = 8$ is 1.860 or more. Since the test statistic is not in this region, this result is not significant, and there is insufficient evidence to reject the null hypothesis that the mean weekly fuel consumption does not change.

(ii) We need to assume that the differences in the fuel consumptions between solar heating and no solar heating are normally distributed.

Exercise 7.3.

(a) *For the data in Example 7.5, perform a χ^2 test for independence, at 5% significance.*
H_0: there is no association between skiing and hiking (events are independent);
H_1: there is an association between skiing and hiking (events are dependent)

	Hiker (O)	Hiker (E)	$\frac{(O-E)^2}{E}$	Non-hiker (O)	Non-hiker (E)	$\frac{(O-E)^2}{E}$
Skiier	178	185	$\frac{(178-185)^2}{185} = 0.264$	45	39	$\frac{(45-39)^2}{39} = 0.923$
Non-skiier	132	125	$\frac{(132-125)^2}{125} = 0.392$	18	26	$\frac{(18-26)^2}{26} = 2.46$

Then $\chi^2 = 0.264 + 0.923 + 0.392 + 2.46 = 4.039$.
The degrees of freedom are $(2-1)(2-1) = 1$, and the critical value for $\nu = 1$ at 5% significance is 3.8415.
Since $4.0339 > 3.841$, there is some evidence of association between preferences for hiking and skiing - these events are dependent, and there is sufficient evidence to reject H_0.

(b) *The following table shows a random sample of 200 customers in a café which only serves goths and emos, and how many ordered tea, coffee or chocolate.*
Use a χ^2 test, at the 5% level of significance, to investigate whether there is any association between the type of drink ordered and music preference.
H_0: there is no association between music type and drink choice (events are independent);
H_1: there is an association between music type and drink choice (events are dependent)

	Tea (O)	*Tea (E)*	$\frac{(O-E)^2}{E}$	*Coffee (O)*	*Coffee (E)*	$\frac{(O-E)^2}{E}$	*Chocolate (O)*	*Chocolate (E)*	$\frac{(O-E)^2}{E}$
Goth	*57*	*46.53*	*2.356*	*26*	*34.31*	*2.013*	*11*	*13.16*	*0.355*
Emo	*42*	*52.47*	*2.089*	*47*	*38.69*	*1.785*	*17*	*14.84*	*0.314*

Here $\nu = (2-1)(3-1) = 1 \times 2 = 2$, *and the critical value for* $\nu = 2$ *is* 5.991.

$$\chi^2 = 2.356 + 2.089 + 2.013 + 1.785 + 0.355 + 0.314 = 8.912$$

Since $8.912 > 5.991$, *there is some evidence of association between music type and choice of drink - these events are dependent, and there is sufficient evidence to reject* H_0.